Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

If $\frac{\pi a}{2}+2(a+b)=L$, a constant, then the function $\frac{\pi a^2}{8}+3ab$ is maximum when $\frac{b}{a}=$

(A) $1+\frac{\pi}{3}$

(B) $1-\frac{\pi}{3}$

(C) $1+\frac{\pi}{6}$

(D) $1-\frac{\pi}{6}$

Q.2

If $A>0, B>0$ and $A+B=\frac{\pi}{3}$, then the maximum value of $\tan A \tan B$ is

(A) $1/\sqrt{3}$

(B) $1/3$

(C) 3

(D) $\sqrt{3}$

Q.3

Slope of tangent to the curve $y=2e^x \sin\left(\frac{\pi}{4}-\frac{x}{2}\right)\cos\left(\frac{\pi}{4}-\frac{x}{2}\right)$ is minimum at $x=$

(A) 0

(B) π

(C) 2π

(D) None of these

Q.4

Let f, g are two continuous and twice derivable functions such that $f(0)=f(3)=0; f(1)\cdot f(2)<0; g(0)=g(3)=0$. Number of roots of equation $f''(x)\cdot g(x)+f'(x)\cdot g'(x)=0$ in $(-1,5)$ cannot be

(A) 2

(B) 3

(C) 4

(D) 5

Q.5

The function $f(x)=\frac{\log(\pi+x)}{\log(e+x)}$, is

(A) increasing on $(0,\infty)$

(B) decreasing on $(0,\infty)$

@IITJEE_Advanced

(C) increasing on $(0, \pi/e)$, decreasing on $(\pi/e, \infty)$

(D) None of the above

Q.6

A curve $y = f(x)$ passes through the point $(4, 3)$ and the normal to the curve at the point happens to be a tangent to the circle $x^2 + y^2 = 25$. The value of $f'(4)$ is

(A) $-3/4$

(B) $3/4$

(C) $4/3$

(D) $-4/3$

Q.7

The angle between tangents at the point of intersection of two curves $x^3 - 3xy^2 + 2 = 0, 3x^2y - y^3 = 2$ is

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

Q.8

Tangents are drawn from origin to the curve $y = \cos x$, their points of contact lie on the curve

(A) $x^2 + y^2 = x^2y^2$

(B) $x^2 - y^2 = xy$

(C) $x^2 + y^2 = 1$

(D) $x^2 - y^2 = x^2y^2$

Q.9

$f(x) = \begin{cases} \sin \frac{\pi x}{2}, 0 \le x < 1 \\ 3 - 2x, \quad x \ge 1 \end{cases}$, then choose the incorrect statements -

(A) $f(x)$ has a local minimum at $x = 1$

(B) $f(x)$ has a local maximum at $x = 1$

(C) $f(x)$ does not have any local maximum or minimum at $x = 1$

(D) $f(x)$ has a global minimum at $x = 1$

Q.10

Let $f(x)$ be monotonic, $f''(x) \ge 0, \lim_{x \to \infty} f(x) = \infty$ and $f\left(2a^2 + a + 1\right) > f\left(3a^2 - 4a + 1\right)$. Possible integral values of a are -

(A) 2

(B) 3

@IITJEE_Advanced

(C) 4

(D) 5

Q.11

Let $f(x) = \sin^4 x + \cos^4 x, g(x) = \sin^6 x + \cos^6 x$ then $-$

(A) whenever $f(x)$ is increasing, $g(x)$ is decreasing

(B) whenever $f(x)$ is decreasing, $g(x)$ is increasing

(C) whenever $f(x)$ is increasing, $g(x)$ is increasing

(D) whenever $f(x)$ is decreasing, $g(x)$ is decreasing

Q.12

Passage

$a(t)$ is a function of t such that $\frac{da}{dt} = 2$ for all values of t and $a = 0$ when $t = 0$.

Further $y = m(t)x + c(t)$ is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0 . Then -

Question:

If the rate of change of distance of vertex of $y = x^2 - 2ax + a^2 + a$ from the origin with respect to t is k, then $k =$

(A) 2

(B) $2\sqrt{2}$

(C) $\sqrt{2}$

(D) $4\sqrt{2}$

Q.13

Passage

$a(t)$ is a function of t such that $\frac{da}{dt} = 2$ for all values of t and $a = 0$ when $t = 0$.

Further $y = m(t)x + c(t)$ is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0 . Then -

Question:

If the rate of change of $c(t)$ with respect to t, when $t = k$ is ℓ, then $\ell =$

(A) $16\sqrt{2} - 2$

(B) $8\sqrt{2} + 2$

(C) $10\sqrt{2} + 2$

(D) $16\sqrt{2} + 2$

Q.14

Passage

$a(t)$ is a function of t such that $\frac{da}{dt} = 2$ for all values of t and $a = 0$ when $t = 0$.

Further $y = m(t)x + c(t)$ is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0 . Then -

Question:

The rate of change of $m(t)$, with respect to t, at $t = \ell$ is -

(A) -2

(B) 2

@IITJEE_Advanced

(C) -4

(D) 4

Q.15

Passage

Let f and g are two functions such that $\mathrm{f}(\mathrm{x})$ and $\mathrm{g}(\mathrm{x})$ are continuous in $[\mathrm{a}, \mathrm{b}]$ and differentiable in (a, b). Then at least one $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$

(i) If $f(a) = f(b)$, then $f'(c) = 0$ (RMVT)

(ii) If $\mathrm{f}(\mathrm{a}) \neq \mathrm{f}$ (b) and $\mathrm{a} \neq \mathrm{b}$, (LMVT)

(iii) If $g'(x) \neq 0$, then $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$ (cauchy theorem)

Question:

The set of values of k_1, for which equation $x^3 - 3x + k = 0$ has two distinct roots in $(0, 1)$ is -

(A) $(1, 4)$

(B) $(0, \infty)$

(C) $(0, 1)$

(D) ϕ

Q.16

Passage

Let f and g are two functions such that $\mathrm{f}(\mathrm{x})$ and $\mathrm{g}(\mathrm{x})$ are continuous in $[\mathrm{a}, \mathrm{b}]$ and differentiable in (a, b). Then at least one $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$

(i) If $f(a) = f(b)$, then $f'(c) = 0$ (RMVT)

(ii) If $f(a) \neq f(b)$ and $a \neq b$, (LMVT)

(iii) If $g'(x) \neq 0$, then $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$ (cauchy theorem)

Question:

Which of the following is true -

(A) $\left|\tan^{-1}\mathrm{x} - \tan^{-1}\mathrm{y}\right| \leq |\mathrm{x} - \mathrm{y}| \forall \mathrm{x}, \mathrm{y} \in \mathrm{R}$

(B) $\left|\tan^{-1}\mathrm{x} - \tan^{-1}\mathrm{y}\right| \geq |\mathrm{x} - \mathrm{y}| \forall \mathrm{x}, \mathrm{y} \in \mathrm{R}$

(C) $|\sin\mathrm{x} - \sin\mathrm{y}| \geq |\mathrm{x} - \mathrm{y}| \forall \mathrm{x}, \mathrm{y} \in \mathrm{R}$

(D) None of these

Q.17

Passage

Let f and g are two functions such that $\mathrm{f}(\mathrm{x})$ and $\mathrm{g}(\mathrm{x})$ are continuous in $[\mathrm{a}, \mathrm{b}]$ and differentiable in (a, b). Then at least one $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$

(i) If $f(a) = f(b)$, then $f'(c) = 0$ (RMVT)

(ii) If $f(a) \neq f(b)$ and $a \neq b$, (LMVT)

(iii) If $g'(x) \neq 0$, then $\frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$ (cauchy theorem)

Question:

Let $0 < \alpha < \theta < \beta < \pi/2$, then $\frac{\sin\alpha - \sin\beta}{\cos\alpha - \cos\beta}$ is equal to -

(A) $\tan\alpha$

(B) $-\tan\theta$

@IITJEE_Advanced

(C) $\cot\theta$

(D) $-\cot\theta$

Q.18

Match the column.

Column I	**Column II**
(A) If x is real, then the greatest and least value of the expression $\frac{x+2}{2x^2+3x+6}$ is	(p) 3
(B) If $a+b=1$, $a>0$, $b>0$, then the minimum value of $\sqrt{\left(1+\frac{1}{a}\right)\left(1+\frac{1}{b}\right)}$ is	(q) 1/3
(C) The maximum value attained by $y = 10 - \lvert x - 10 \rvert$, $-1 \le x \le 3$ is	(r) 5
(D) If P $(t^2, 2t)$, $t \in [0, 2]$ is an arbitrary point on parabola $y^2 = 4x$. Q is foot of perpendicular from focus S on the tangent at P, then maximum area of triangle PQS is	(s)–1/13

(A) A-p,r ; B-p ; C-r ; D-q

(B) A-q,s ; B-p ; C-p ; D-r

(C) A-q,s ; B-p ; C-r ; D-q

(D) A-q,p ; B-r ; C-s ; D-q

Q.19

The largest possible value of the expression $y = \sqrt{x-2} + 2\sqrt{3-x}$ is $\sqrt{A}$. Find the value of A.

Q.20

The number of tangents to the curve $y^2 - 2x^3 - 4y + 8 = 0$ which passes through the point $(1, 2)$ is

Answers & Solutions

Q.1 Answer:

$1 + \frac{\pi}{6}$

Solution:

$$\text{(c): Eliminating } b,\ f(a) = \frac{\pi a^2}{8} + \frac{3a}{2}\left[L - \frac{\pi a}{2} - 2a\right]$$

$$= \frac{3aL}{2} - (5\pi + 24)\frac{a^2}{8},\ f'(a) = 0 \Rightarrow a = \frac{6L}{5\pi + 24}$$

$$b = \frac{1}{2}\left[L - \frac{\pi a}{2} - 2a\right] = \frac{(\pi + 6)L}{5\pi + 24},\ \frac{b}{a} = \frac{\pi + 6}{6}$$

Q.2 Answer:

$1/3$

Solution:

@IITJEE_Advanced

We have, $A + B = \pi/3$.
$\therefore B = \frac{\pi}{3} - A$
$\Rightarrow \tan B = \frac{\sqrt{3} - \tan A}{1 + \sqrt{3}\tan A}$
Let $\mathrm{Z} = \tan \mathrm{A} + \tan \mathrm{B}$.
Then $\mathrm{Z} = \tan \mathrm{A} + \frac{\sqrt{3} - \tan \mathrm{A}}{1 + \sqrt{3}\tan \mathrm{A}} = \frac{\sqrt{3}(\tan^2 A + 1)}{1 + \sqrt{3}\tan \mathrm{A}}$
$\Rightarrow \mathrm{z} = \frac{\sqrt{3}(\mathrm{x}^2 + 1)}{1 + \sqrt{3}\mathrm{x}}$, where $\mathrm{x} = \tan \mathrm{A}$
$\Rightarrow \quad \frac{\mathrm{dZ}}{\mathrm{dx}} = \frac{(\mathrm{x} + \sqrt{3})(3\mathrm{x} - \sqrt{3})}{(1 + \sqrt{3}\mathrm{x})^2}$
For $\max \mathrm{Z}$, $\frac{\mathrm{dZ}}{\mathrm{dx}} = 0$
$\Rightarrow \mathrm{x} = \frac{1}{\sqrt{3}}, -\sqrt{3}$
$x \neq -\sqrt{3}$ because $A + B = \frac{\pi}{3}$ which implies that $x = \tan A > 0$ it can be easily checked that $\frac{d^2 Z}{dx^2} < 0$ for $x = \frac{1}{\sqrt{3}}$
Hence, Z is $\max$. for $x = \frac{1}{\sqrt{3}}$ i.e. $\tan A = \frac{1}{\sqrt{3}}$ or $A = \pi/6$.
For this value of x, $Z = 1/3$
Hence, $\tan A \cdot \tan B = \frac{1}{\sqrt{3}}\frac{1}{\sqrt{3}} = \frac{1}{3}$

Q.3 Answer:

π

Solution:

The slope of the tangent to the curve $\mathrm{y} = 2\mathrm{e}^{\mathrm{x}} \sin\left(\frac{\pi}{4} - \frac{\mathrm{x}}{2}\right) \cos\left(\frac{\pi}{4} - \frac{\mathrm{x}}{2}\right) = \mathrm{e}^{\mathrm{x}} \cos \mathrm{x}$ is

$$S = \frac{dy}{dx} = e^x(-\sin x + \cos x)$$
$$\text{Now, } \frac{dS}{dx} = e^x(-\sin x + \cos x - \cos x - \sin x) = -2e^x \sin x$$
$$\frac{dS}{dx} = 0 \Rightarrow -2e^x \sin x \Rightarrow x = 0, \pi, 2\pi \quad (\because 0 \leq x \leq 2\pi)$$

Value of S at $x = 0$ is 1 , value of S at $x = \pi$ is $-e^{\pi}$, value of S at $x = 2\pi$ is $e^{2\pi}$
$\therefore$ S is minimum at $\mathrm{x} = \pi$.

Q.4 Answer:

2

Solution:

Min. no.of roots of $f(x) = 0$ is 3 .
Min. no. of roots of $\mathrm{g(x)} = 0$ is 2 .
Given equation is $(f'(x) \cdot g(x))' = 0$
$\because f'(x) = 0$ has minimum 2 roots
$\therefore f'(\mathrm{x}) \cdot \mathrm{g(x)} = 0$ has minimum 4 roots.
$\Rightarrow (f'(\mathrm{x}) \cdot \mathrm{g(x)})' = 0$ has minimum 3 roots.

Q.5 Answer:

decreasing on $(0, \infty)$

Solution:

@IITJEE_Advanced

Given, $f(x) = \frac{\log(\pi+x)}{\log(e+x)} \Rightarrow \frac{\log(e+x)\times\frac{1}{\pi+x} - \log(\pi+x)\times\frac{1}{e+x}}{[\log(e+x)]^2}$

$\Rightarrow \frac{(e+x)\log(e+x) - (\pi+x)\log(\pi+x)}{(e+x)(\pi+x)[\log(e+x)]^2}$

$\Rightarrow \frac{x\log\left(\frac{e+x}{\pi+x}\right) - [e\log(e+x) - \pi\log(\pi+x)]}{(e+x)(\pi+x)[\log(e+x)]^2}$

Now, $\pi + \mathrm{x} > \mathrm{e} + \mathrm{x} \Rightarrow \log\left(\frac{\mathrm{e}+\mathrm{x}}{\pi+\mathrm{x}}\right) < 0$ and also elog $(e + x) < \pi\log(\pi + x) \Rightarrow f'(x) < 0$

Thus, $\mathrm{f}(\mathrm{x})$ is decreasing on $(0, \infty)$

Q.6 Answer:

$3/4$

Solution:

As $(4, 3)$ lies on the circle, the normal to

$\mathrm{y} = \mathrm{f}(\mathrm{x})$ is the tangent to the circle $(4, 3)$ so that the two curves intersect orthogonally at $(4, 3)$.

$\mathrm{f}'(4) =$ the slope of $\mathrm{y} = \mathrm{f}(\mathrm{x})$ at $(4, 3) = -$ the reciprocal of the slope of the circle $(4, 3) = \frac{3}{4}$.

Q.7 Answer:

$\frac{\pi}{2}$

Solution:

Let the point of intersection is (x, y)

$$x^3 - 3xy^2 + 2 = 0 \Rightarrow \frac{dy}{dx} = \frac{x^2 - y^2}{2xy}, \quad 3x^2y - y^3 = 2 \Rightarrow \frac{dy}{dx} = \frac{2xy}{y^2 - x^2}, \quad m_1, m_2 = -1 \Rightarrow \theta = 90^0$$

Q.8 Answer:

$x^2 - y^2 = x^2y^2$

Solution:

Let $(\mathrm{x}_1, \mathrm{y}_1)$ be one of the points of contact. Given curve is $\mathrm{y} = \cos\mathrm{x}$

$$\Rightarrow \frac{dy}{dx} = -\sin x$$

$$\Rightarrow \left|\frac{dy}{dx}\right|_{(x_1, y_1)} = -\sin x_1$$

Now the equation of the tangent at $(\mathrm{x}_1, \mathrm{y}_1)$ is

$$\mathrm{y} - \mathrm{y}_1 = \left(\frac{\mathrm{dy}}{\mathrm{dx}}\right)_{(\mathrm{x}_1, \mathrm{y}_1)} (\mathrm{x} - \mathrm{x}_1)$$

$$\Rightarrow \mathrm{y} - \mathrm{y}_1 = -\sin\mathrm{x}_1\,(\mathrm{x} - \mathrm{x}_1)$$

Since, it is given that equation of tangent passes through origin

$$\Rightarrow 0 - \mathrm{y}_1 = -\sin\mathrm{x}_1\,(0 - \mathrm{x}_1)$$

$$\therefore \mathrm{y}_1 = -\mathrm{x}_1\sin\mathrm{x}_1 \ldots (\mathrm{i})$$

also, point $(\mathrm{x}_1, \mathrm{y}_1)$ lies on $\therefore \mathrm{y}_1 = \cos\mathrm{x}_1$

From Eqs (i), (ii) we get

@IITJEE_Advanced

$\sin^2 x_1 + \cos^2 x_1 = \frac{y_1^2}{x_1^2} + y_1^2 = 1$

$\Rightarrow x_1^2 = y_1^2 + y_1^2 x_1^2$

Hence, the locus of $x^2 = y^2 + y^2x^2 \Rightarrow x^2y^2 = x^2 - y^2$

Q.9 Answer:

f (x) has a local minimum at $x = 1$

Solution:

$f'(x) = \frac{\pi}{2}\cos\frac{\pi}{2}x, 0 \le x < 1 = -2, x \ge 1$

$\therefore$ $f'(x)$ **changes sign from positive to negative from the left side of** $x = 1$ **to the right side of** $x = 1$.

f **(x) changes from an increasing function to a decreasing function.**

$\therefore$ $f(x)$ **has a local maximum at** $x = 1$.

Q.10 Answer:

2

Solution:

$f(x)$ **is monotonic increasing**

$\Rightarrow f\left(2a^2 + a + 1\right) > f\left(3a^2 - 4a + 1\right)$

$\Rightarrow 2a^2 + a + 1 > 3a^2 - 4a + 1$

$-a^2 + 5a > 0$

$a^2 - 5a < 0$

$0 < a < 5$

Q.11 Answer:

whenever $f(x)$ is increasing, $g(x)$ is increasing

Solution:

$f(x) = \sin^4 x + \cos^4 x$

$f(x) = \left(\sin^2 x + \cos^2 x\right)^2 - 2\sin^2 x\cos^2 x$

$f(x) = 1 - 2\sin^2 x\cos^2 x$

$f(x) = 1 - \frac{\sin^2 2x}{2}$

$f'(x) = \frac{-2sin(2x)\cdot cos2x\cdot 2}{2}$

$f'(x) = -\sin 4x$

$g(x) = \left(\sin^2 x + \cos^2 x\right)\left(\sin^4 x + \cos^4 x - \sin^2 x\cos^2 x\right)$

$g(x) = 1 - 3\sin^2 x\cos^2 x$

$g(x) = 1 - \frac{3}{4}\sin^2 2x$

$g'(x) = -\frac{3}{2}\sin 4x$

$g'(x) = 1$

It Can be easily said that the sign of $f'(x)$ **and** $g'(x)$ **are same,**

So, whenever $f(x)$ **is increasing,** $g(x)$ **is inceasing and when ever** $f(x)$ **is decreasing,** $g(x)$ **is decreasing option (3),** (4) **are correct**

Q.12 Answer:

$2\sqrt{2}$

Solution:

@IITJEE_Advanced

$$\frac{da}{dt} = 2 \Rightarrow a = 2t + c$$

$\because c = 0 \{\because a = 0, \text{ when } t = 0\}$

$\therefore$ t he curve $y = x^2 - 2ax + a^2 + a$ becomes

$$y = x^2 - 4tx + 4t^2 + 2t$$

if $x = 0$, then $\mathbf{y} = 4t^2 + 2t$

$$\frac{dy}{dx} = 2x - 4t$$

$$\therefore \left.\frac{dy}{dx}\right|_{atx=0} = -4t$$

$\therefore$ Equation of the tangent $y - \left(4t^2 + 2t\right) = -4t(x - 0)$ i.e., $y = -4tx + 4t^2 + 2t$

Vertex of $y = x^2 - 4tx + 4t^2 + 2t$ is $(2t, 2t)$

$\therefore$ Distance of vertex from the origin $= 2\sqrt{2}t$

$\therefore$ Rate of change of distance of vertex from origin with respect to $t = 2\sqrt{2}$

i.e. $k = 2\sqrt{2}$

$$c(t) = 4t^2 + 2t$$

$$\therefore \frac{dc}{dt} = 8t + 2$$

$$\therefore \left.\frac{dc}{dt}\right|_{att=2\sqrt{2}} = 16\sqrt{2} + 2$$

$\therefore \ell = 16\sqrt{2} + 2$

$$m(t) = -4t$$

$$\therefore \frac{dm}{dt} = -4$$

$$\therefore \left.\frac{dm}{dt}\right|_{att=\ell} = -4$$

Q.13 Answer:

$16\sqrt{2} + 2$

Solution:

@IITJEE_Advanced

$$\frac{da}{dt} = 2 \Rightarrow a = 2t + c$$

$\because c = 0 \{\because a = 0, \text{ when } t = 0\}$

$\therefore$ t he curve $y = x^2 - 2ax + a^2 + a$ becomes

$$y = x^2 - 4tx + 4t^2 + 2t$$

if $x = 0$, then $y = 4t^2 + 2t$

$$\frac{dy}{dx} = 2x - 4t$$

$$\therefore \left.\frac{dy}{dx}\right|_{atx=0} = -4t$$

$\therefore$ Equation of the tangent $y - \left(4t^2 + 2t\right) = -4t(x - 0)$ i.e., $y = -4tx + 4t^2 + 2t$

Vertex of $y = x^2 - 4tx + 4t^2 + 2t$ is $(2t, 2t)$

$\therefore$ Distance of vertex from the origin $= 2\sqrt{2}t$

$\therefore$ Rate of change of distance of vertex from origin with respect to $t = 2\sqrt{2}$

i.e. $k = 2\sqrt{2}$

$$c(t) = 4t^2 + 2t$$

$$\therefore \frac{dc}{dt} = 8t + 2$$

$$\therefore \left.\frac{dc}{dt}\right|_{att=2\sqrt{2}} = 16\sqrt{2} + 2$$

$\therefore \ell = 16\sqrt{2} + 2$

$$m(t) = -4t$$

$$\therefore \frac{dm}{dt} = -4$$

$$\therefore \left.\frac{dm}{dt}\right|_{att=\ell} = -4$$

Q.14 Answer:

-4

Solution:

@IITJEE_Advanced

$\frac{da}{dt} = 2 \Rightarrow a = 2t + c$
$\because c = 0\{\because a = 0,$ when $t = 0\}$
$\therefore$ t he curve $y = x^2 - 2ax + a^2 + a$ becomes
$y = x^2 - 4tx + 4t^2 + 2t$
if $x = 0,$ then $\mathbf{y} = 4t^2 + 2t$
$\frac{dy}{dx} = 2x - 4t$
$\therefore \left.\frac{dy}{dx}\right|_{atx=0} = -4t$
$\therefore$ Equation of the tangent $y - \left(4t^2 + 2t\right) = -4t(x - 0)$ i.e., $y = -4tx + 4t^2 + 2t$
Vertex of $y = x^2 - 4tx + 4t^2 + 2t$ is $(2t, 2t)$
$\therefore$ Distance of vertex from the origin $= 2\sqrt{2}t$
$\therefore$ Rate of change of distance of vertex from origin with respect to $t = 2\sqrt{2}$
i.e. $k = 2\sqrt{2}$
$c(t) = 4t^2 + 2t$
$\therefore \frac{dc}{dt} = 8t + 2$
$\therefore \left.\frac{dc}{dt}\right|_{att=2\sqrt{2}} = 16\sqrt{2} + 2$
$\therefore \quad \ell = 16\sqrt{2} + 2$
$m(t) = -4t$
$\therefore \quad \frac{dm}{dt} = -4$
$\therefore \left.\frac{dm}{dt}\right|_{att=\ell} = -4$

Q.15 Answer:

ϕ

Solution:

Let $0 < \alpha < \beta < 1$, and α, β are the roots of
$f(x) = x^3 - 3x + k = 0 \Rightarrow f(\alpha) = f(\beta) = 0$
$\Rightarrow$f(x) satisfies RMVT $\Rightarrow f'(c) = 0 \Rightarrow 3c^2 = 3 \Rightarrow c = \pm 1$
but c must be lies between α and β. Hence $k \in \phi$.

Q.16 Answer:

$\left|\tan^{-1} x - \tan^{-1} y\right| \leq |x - y| \forall x, y \in R$

Solution:

Let $f(x) = \tan^{-1}(x)$ then for some $\alpha \in (x, y), f'(\alpha) = \frac{\tan^{-1} y - \tan^{-1} x}{y - x}$ (LMVT)
$\Rightarrow \left|\frac{1}{1+\alpha^2}\right| = \left|\frac{\tan^{-1} x - \tan^{-1} y}{x - y}\right| \quad \left(\left|\frac{1}{1+\alpha^2}\right| \leq 1\right) \Rightarrow \left|\tan^{-1} x - \tan^{-1} y\right| \leq |x - y|$

Q.17 Answer:

$-\cot\theta$

Solution:

Let $f(x) = \sin x$ and $g(x) = \cos x$.
also $\sin x \neq 0$ for $x \in \left(0, \frac{\pi}{2}\right)$, then by cauchy's theorem
$\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} = \frac{f'(\theta)}{g'(\theta)} \Rightarrow \frac{\sin\beta - \sin\alpha}{\cos\beta - \cos\alpha} = \frac{\cos\theta}{-\sin\theta} = -\cot\theta$

Q.18 Answer:

A-q,s ; B-p ; C-p ; D-r

Solution:

@IITJEE_Advanced

(A) → qs ; (B) → p ; (C) → p ; (D) → r

(A) $y = \frac{x+2}{2x^2+3x+6}$

$2yx^2 + 3xy + 6y = x + 2$

$\Rightarrow 2yx^2 + x(3y - 1) + 6y - 2 = 0$

$D \geq 0$

$\Rightarrow \quad (3y - 1)^2 - 8y(6y - 2) \geq 0$

$9y^2 - 6y + 1 - 48y^2 + 16y \geq 0$

$\Rightarrow -39y2 + 10y + 1 \geq 0$

$39y^2 - 10y - 1 \leq 0$

$(13y + 1)(3y - 1) \leq 0$

$\frac{-1}{13} \leq y \leq \frac{1}{3}$

(B) $a + b = 1$

$\sqrt{\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)} = \sqrt{1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{ab}} = \sqrt{1 + \frac{2}{ab}}$

$\sqrt{ab} < \frac{a+b}{2} = \frac{1}{2} \quad \therefore ab < \frac{1}{4} \Rightarrow \frac{1}{ab} > 4$

$\therefore \sqrt{\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)} \geq \sqrt{1 + 8} = 3$

(C) $y = 10 - |x - 10|$

$-1 \leq x \leq 3$

$-11 \leq x - 10 \leq -7$

$\Rightarrow \quad 7 \leq |x - 10| \leq 11$

$\Rightarrow -1 \leq y \leq 3 \therefore \quad y = 10 - (10 - x) = x$

$\therefore$ **maximum value of** $y = 3$

(D) Equation of tangent at P **is** $ty = x + t^2$ **it intersects the line** $x = 0$ **at** Q

$\therefore$ **coordinates of** Q **are** $(0, t)$

$\therefore$ **area of** $\Delta PQS = \frac{1}{2}\begin{vmatrix} 0 & t & 1 \\ 1 & 0 & 1 \\ t^2 & 2t & 1 \end{vmatrix} = \frac{1}{2}\left[-t\left(1 - t^2\right) + 2t\right] = \frac{1}{2}\left(t + t^3\right) \frac{dA}{dt} = \frac{1}{2}\left(3t^2 + 1\right) >$

$0 \ \forall t \in [0, 2]$

$\therefore$ **Area is maximum for** $t = 2$ **max area** $= \frac{1}{2}[2 + 8] = 5$

Q.19 Answer:

—

Solution:

Domain is $[2, 3]$

Now $\frac{dy}{dx} = \frac{1}{2\sqrt{x-2}} - \frac{1}{\sqrt{3-x}}$

for maximum or minimum $\frac{dy}{dx} = 0$

$4(x - 2) = 3 - x \Rightarrow 5x = 11 \Rightarrow x = \frac{11}{5}$

Now, $f(2) = 2, f(3) = 1, f\left(\frac{11}{5}\right) = \frac{1}{\sqrt{5}} + 2 \cdot \frac{2}{\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5}$

Q.20 Answer:

—

Solution:

$\frac{dy}{dx} = \frac{6x^2}{2y-4} = \frac{3x^2}{y-2}$

Equation of tangent at (x_1, y_1) **is** $3x_1^2x - 3x_1^3 = yy_1 - 2y - y_1^2 + 2y_1$

But it passes through $(1, 2)$

@IITJEE_Advanced

$\therefore 3x_1^3 - 3x_1^2 - y_1^2 + 4y_1 - 4 = 0$

But, $y_1^2 - 4y_1 = 2x_1^3 - 8$

Find (1) & (2), $x_1^3 - 3x_1^2 + 4 = 0$

$\Rightarrow x_1 = -1(\text{ or }) 2$

At $x_1 = -1 \Rightarrow' y_1'$ **imaginary**

At $x_1 = 2 \Rightarrow y_1^2 - 4y_1 - 8 = 0$

Hence, there exist 2 points

@IITJEE_Advanced